

# AI Ethics Assignment 1:

Responsible AI Principles and the Model Development Life Cycle

**Prepared by:**

Atikulrehman Patel

Neil McGuffin

Roy Aggarwal

William Makino

**Course:** ETHI-401-IAI

**Instructor:** Steven Nich

**Date:** February 27, 2026

# Introduction

This paper applies four responsible AI principles—Accountability, Reliability & Safety, Privacy & Security, and Fairness—across the CRISP-DM model development lifecycle. Our use case is a street-level traffic optimization system modeled on Google Street View, where vehicles collect imagery and sensor data to generate routing recommendations that reduce congestion.

## Definitions

### Accountability

Bovens defines accountability as a structured relationship in which an actor must justify decisions to a forum with authority to question and sanction (Bovens, 2007). Effective accountability requires clear decision authority, enforceability, and responsiveness. In AI governance, accountability extends beyond internal responsibility to include regulatory and public oversight (Ammanath, 2022; European Union, 2024). It is distinct from responsibility (task ownership) and liability (legal consequences). In complex AI systems, unclear boundaries between these concepts can obscure decision authority and weaken the obligation to justify outcomes (Bovens, 2007; Ammanath, 2022).

### **Reliability + Safety**

Reliability in AI refers to consistent, dependable performance under varying conditions, including resilience to noise, input degradation, and model drift (Ammanath, 2022; NIST, n.d.). It addresses whether the system functions as intended with measurable stability. Safety, by contrast, concerns harm prevention. It requires defining acceptable risk thresholds, identifying foreseeable harms, and establishing constraints, guardrails, and oversight mechanisms. While safety depends on reliability, technical performance alone does not guarantee harm mitigation (Ammanath, 2022; NIST, n.d.).

## **Privacy + Security**

Privacy concerns lawful and limited data collection, purpose limitation, minimization of personally identifiable information, and governance controls to prevent misuse (Ammanath, 2022). It focuses on protecting individuals from harms arising from data use and retention. Security protects systems, data, and model artifacts from unauthorized access, tampering, or adversarial compromise (NIST, n.d.). The two are interdependent: weak security can expose private data, while excessive data collection increases the impact of any security failure.

## **Fairness**

In Ammanath's framework, fairness requires identifying and mitigating bias, ensuring equitable treatment across groups, and preventing disproportionate harm, while recognizing that fairness is context-dependent and ongoing (Ammanath, 2022). It operates at both the statistical level (model performance across groups) and the institutional level (how systems interact with existing social inequities). Technical corrections alone are insufficient without broader governance consideration.

## **Business Understanding**

The business objective of this system is congestion mitigation through advisory recommendations provided to individual drivers that improve traffic flow and driver quality of life by minimizing or avoiding gridlock. Street-level imagery vehicles serve as a data acquisition mechanism supporting this objective.

In the initial phase of data collection, street-level vehicles will serve as image and sensor data acquisition platforms. These will provide comprehensive mapping of global neighborhoods and road networks. However, from an ethical standpoint, the primary constraint is not technical. It reflects the principle of non-maleficence: the deployment of these vehicles must not cause harm to motorists, pedestrians, cyclists, or property (Ammanath, 2022). Reliability in this phase is defined as the system's ability to perform consistently under varying environmental conditions (for

example, rain, glare, or heavy traffic) without failure. A failure in reliability—such as a sensor glitch—may propagate into downstream safety risks once processed by the model. Therefore, our business success criteria must include a zero-incident safety record alongside data quality metrics.

Alongside data quality, data security and public privacy must also be built into the project objective from the start, and not treated as afterthoughts (Ammanath, 2022). Our problem statement should explicitly frame the goal as characterizing infrastructure only, with stated non-goals such as no identification of individuals, no persistent vehicle tracking, and no inferences about residents, which is consistent with the purpose-limitation principle both Google and Microsoft embed in their responsible AI commitments (Google, 2026; Microsoft Responsible AI, 2026). At this stage we also map applicable regulations and data sensitivities, since street imagery will also capture faces, license plates, and private spaces, creating legal and ethical exposure if the scope is not tightly defined (Mehan, 2022).

For this use case (traffic flow optimization using street-level imagery), several questions must be foregrounded at this stage of CRISP-DM pertaining to stakeholders, risks, oversight, and authority regarding the data, the model development, and the eventual model usage. Specific accountability-related questions include: who defines acceptable imagery standards? Who signs off on model accuracy? Who has authority to halt deployment? (Ammanath, 2022).

While improving online navigation and traffic patterns, in the early stages we should work to determine the extent to which systematic image capture and mapping may skew against some groups of people.

The OECD (2019) states that AI systems should promote inclusive growth, sustainable development, and well-being (p. 3). As such, the analysis of fairness at this step must be conducted by considering the benefits and possible burdens that our use case might impose.

Post-hoc review should not be the default; embedding these principles early establishes preventative governance and escalation pathways before harm occurs (NIST, n.d.).

## Data Understanding

Geburu et al.'s (2021) Datasheets for Datasets framework provides a foundation for accountability in Data Understanding. Documenting dataset purpose, scope, creators, affiliations, and funding clarifies decision authority and minimizes diffusion of responsibility.

The data in this context is twofold: the navigational data collected by the car and the photographic data used for the mapping/routing. Navigational Data such as LiDAR, Radar, and GPS inputs must be verified for accuracy. Ethics enters here through the lens of Data Integrity. If the sensors generate erroneous readings, or fail, the data becomes unreliable, potentially introducing safety risks downstream (NIST, n.d.).

Safety implications require acknowledging "edge cases," such as unmapped construction zones or erratic human behavior, which represent the highest risk to vehicle safety (NIST, n.d.).

Early Google Street View reports emphasized the need to assess the sensitivity of raw data (Federal Communications Commission [FCC], 2012). In our use case, the same rigor must be observed. Street-level imagery and sensor logs can reveal far more than road geometry. They capture identifiable people, vehicles, and behavioral patterns that could imply where someone lives or when a property is unoccupied (Mehan, 2022). Each data element should be classified by privacy risk, with raw imagery and timestamps flagged as high-risk and purely geometric features flagged as low-risk. The potential for re-identification through data combination must be explicitly documented (Ammanath, 2022; Loukides, Mason, & Patil, 2018).

Governance risks are brought about by large scale image collection. As an example, in one of the investigations into the activities of Google Street View, the Federal Communications Commission discovered that the internal controls used by Google were inadequate as there was no organized supervision to some of the data practices (FCC, 2012, p. 5).

According to NIST (2023), unreliable or unbalanced datasets may lead to dangerous bias (p. 10). When certain communities are not adequately represented in mapping updates, digital inequality could occur.

Barocas et al. (2019) also describe that representational harms may be experienced in cases where particular groups are disproportionately exposed or not included in datasets (p. 23). One could foresee a situation in our use case where traffic is inadvertently routed down side streets or through lower income neighborhoods where vehicle traffic density is lower, but pedestrian activity is elevated, thus exposing the local population to more traffic and thus, more potential for incidents involving vehicles than expected.

## **Data Preparation**

In this stage of the CRISP-DM framework, we look at accountability in Data Preparation. Here, accountability shifts to: Who defines and approves feature inclusion, transformations, and derived features, and what governance and escalation pathways exist if risk concerns arise? (Gebru et al., 2021)

We are ensuring engineered features are reviewed under established authority structures that we identified during the Business Understanding phase, and that inclusion decisions are documented and justified.

We also reassess accountability mechanisms as new actors engage more fully with the project (feature engineers, modelers) and ensure that the accountability framework is clearly communicated and still valid. If it is not communicated, the resulting ambiguity may result in inadvertent diffusion of accountability and authority that we aim to prevent. If the accountability framework is no longer valid, it must be addressed to avoid similar issues.

In traditional CRISP-DM, this phase involves cleaning and transforming data. For our use case, this extends to the real-time processing pipeline. Reliability is ensured by redundancy: Using multiple data streams to cross-verify the initial surroundings, and later to validate the model's

outputs; and noise reduction: Filtering out sensor artifacts and environmental effects that could contaminate the data, such as burnt pixels, heavy snow, or dust.

From an ethical perspective, the preparation of data must prioritize representative sampling to ensure the vehicle recognizes diverse factors across different neighborhoods, including things like the type and density of pedestrian mobility in the area, the amount of parked traffic, signage, etc.

This phase is where privacy protections become technically concrete. Faces and license plates must be irreversibly blurred before data enters any pipeline, and views of private spaces should be filtered out (Google, 2026). Raw imagery should then be converted into abstract, numerical features such as road width, pothole severity index, sidewalk continuity score, and parking occupancy ratio, after which the underlying images should be deleted or tightly access-controlled so that the model does not depend on identifiable data (Microsoft Responsible AI, 2026; Ammanath, 2022). Encryption in transit and at rest, strict access logging, and defined data retention limits are foundational controls at this stage (Mehan, 2022; Loukides, Mason, & Patil, 2018).

Data preparation is a critical stage in the AI lifecycle. It is where the threat of unfairness tends to become more highly manifest. As with Google Street View, our use case must contend with the issues of bias and representation due to the volume of data, including image data, collected.

As noted previously, the quality of the datasets contributes to fairness. The National Institute of Standards and Technology (2023) states that “unreliable or unbalanced datasets may lead to dangerous bias” (p. 10). If certain communities are underrepresented in our data, and the data is not sufficiently vetted or cleaned, then that disparity could become anchored in the modeling, digital inequality may result.

## **Modeling**

For the data modeling/tuning/testing cycle, we've agreed to adopt a proposed variation of the model cards suggested by Timnit Gebru et al. in their Model Cards for Models paper. While the adoption as outlined in the paper leans more towards compliance oversight, transparency, explainability, and fairness in the model, the questions raised during the process of formulating what these model cards will contain (as per the paper) requires documentation of modeling decisions. Documentation makes the decision authority visible, and thus accountable.

As an example, section 4.4 in Gebru et al. (Model cards) on Metrics asks about the measures being reported, why they were selected, whether decision thresholds are used and why, and how estimates are calculated within the model. (Gebru et al., 2019).

This phase determines how routing decisions are generated for users. The model must be trained for Predictive Reliability. This is accomplished through Constraint-Based Modeling. The AI's "policy" must be explicitly encoded with safety buffers that prioritize human life over traffic flow efficiency. For instance, if a neighborhood is too congested for a vehicle to navigate safely, the model should trigger a "Safe State" preventing the route from being recommended as an option, even if that means traffic slowdown occurs elsewhere.

The model should be trained on sanitized, feature-level data rather than raw imagery, ensuring it learns patterns about infrastructure and not about specific people or vehicles (Loukides, Mason, & Patil, 2018). Where feasible, processing should happen at the edge (on the vehicle itself) so sensitive frames never leave the car (Google, 2026; Microsoft Responsible AI, 2026). Risks of model memorization and inversion must also be addressed through regularization and label granularity constraints, and the model must be robust against adversarial manipulation such as tampered parking signs or painted road markings (Ammanath, 2022; Mehan, 2022).

Computer vision models are applied to face detection and license plates blurring in Google Street View. A similar approach is proposed for our use case. The issue of fairness arises when we consider whether these procedures are applied unevenly towards demographic groups.



A study conducted by Joy Buolamwini and Timnit Gebru (2018) revealed that the error rates varied significantly between gender and skin type (p. 11). This shows that the performance results of visual recognition systems can be disproportional and this can have unfair effects.

According to the National Institute of Standards and Technology (2023), AI systems are expected to be tested with regard to the differences in their performance across demographic groups (p. 12). In our use case, this would necessarily extend to the various neighborhoods or regions that are dominated by a particular demographic. This supports the notion that the modeling of fairness involves systematic assessment of elements other than total accuracy in the model.

Moreover, the model needs to be continually monitored post-deployment. According to NIST (2023), AI risk management is supposed to be an iterative process (p. 18). Unless a continuous assessment is conducted for things like bias and fairness, the differences in performance might remain unchanged or even increase over time.

## Evaluation

When deploying a public-facing routing model, the CRISP-DM Evaluation phase must extend beyond business performance metrics to include structured risk management. Frameworks such as the NIST AI Risk Management Framework (RMF) and the EU AI Act guide this process. Independent validation is essential: testing must be separated from development to mitigate confirmation bias and conflicts of interest. Accountability therefore requires clear oversight of metric selection, performance thresholds, risk sign-offs, escalation pathways, and regulatory compliance. Because the model provides public routing advice, it must align with municipal traffic laws and broader transportation legislation, with documented limitations and residual risks. Evaluation must go beyond accuracy metrics and include explicit privacy and security testing, especially when systems draw on large, linkable public-sector datasets (Princeton University Center for Human Values & Princeton Center for Information Technology Policy [UCHV & CITP], 2018; Ammanath, 2022). This includes testing whether outputs and derived features could enable

sensitive inferences when combined with other data, particularly where features reflect human behavior, such as parking occupancy, which is inherently behavioral (Ammanath, 2022; Loukides, Mason, & Patil, 2018). Adversarial stress tests, re-identification attempts, and API misuse scenarios should all be part of the evaluation suite, and if any of these reveal unacceptable risk, the design must be revised before deployment even if predictive performance is strong (Microsoft Responsible AI, 2026; Loukides, Mason, & Patil, 2018).

Before issuing public routing recommendations, the model must be evaluated against rigorous safety, reliability, and fairness benchmarks. While CRISP-DM traditionally assesses alignment with business objectives, AI governance expands the Evaluation phase to include robustness testing, adversarial analysis, and statistical validation to ensure routing improvements are genuine and resilient to manipulation (e.g., vandalism, anomalous traffic clustering, or unexpected incidents). Pre-deployment validation must also address procedural fairness through standardized testing protocols and subgroup performance analysis.

As emphasized by the NIST AI Risk Management Framework (2023), continuous evaluation and monitoring are central to managing AI risk. Public-facing systems can scale rapidly, and without structured oversight, reliability failures may evolve into systemic fairness risks (Barocas et al., 2019). Accordingly, accountability requires transparent documentation, ongoing performance review, and clearly defined monitoring processes consistent with OECD (2019) principles of transparency and responsible governance.

## Deployment

The final phase of the CRISP-DM framework, Deployment, requires transitioning from development-focused evaluation to continuous, real-world risk management. As the system interacts with the public, accountability expands to include broader societal stakeholders and new legislative requirements. Proactive measures must be established to ensure the model remains reliable, safe, secure, and fair as it encounters dynamic environments.

For the final layer of CRISP-DM, all accountability mechanisms must remain forward-facing and proactive. Without this, deployment accountability degrades into reactive blame assignment (Ammanath, 2022). Once deployed, the forum expands (Bovens, 2007), requiring actors with specialized knowledge across legislative, judicial, and advocacy domains. The EU AI Act exemplifies the breadth of consideration required once an AI system interacts with the public. Consequently, all accountability strategies from Ammanath, Bovens, NIST, Gebru, and the EU AI Act must be continually assessed as the system matures, maintaining clarity of authority throughout its operational life, not just during development.

The final phase involves the rollout into traffic networks. Reliability and safety are maintained here through Human-in-the-loop (HITL) mechanisms, where drivers and others monitor these systems to provide a necessary safety net for AI uncertainty. Incident reporting is also vital; any "disengagement" (where the AI fails and the human takes over) is fed back into the Business Understanding phase to restart the CRISP-DM cycle.

Outputs should be surfaced to users at a level of granularity that is useful for navigation but too coarse to enable surveillance or enforcement, such as expressing parking guidance as historical trends rather than live spot availability (Google, 2026; Mehan, 2022). Internal access to fine-grained data must be governed through authentication, rate limiting, and access controls to prevent misuse, such as discriminatory routing or punitive applications (Ammanath, 2022; Microsoft Responsible AI, 2026). A clear retention policy should ensure sensitive data is purged once it no longer serves a routing or accessibility purpose, and a feedback channel for community concerns should be built in from launch (Loukides, Mason, & Patil, 2018).

Fairness and accountability extend beyond initial deployment; consistent with NIST's iterative risk management model and Ammanath's lifecycle framing, ongoing monitoring and stakeholder feedback are required to sustain trust.

## Opportunities and Challenges

Applying responsible AI principles across the CRISP-DM lifecycle presents both structural challenges and opportunities. Challenges include balancing data utility with privacy, preventing bias and representational harms in large-scale imagery collection, maintaining independent validation, and managing regulatory compliance. These constraints can increase development complexity and cost. (NIST, 2023; Ammanath, 2022)

However, structured frameworks such as the NIST AI Risk Management Framework, model cards and datasheets, and regulatory guidance under the EU AI Act provide actionable governance scaffolding. External audit processes, red-teaming practices, and stakeholder engagement mechanisms can further strengthen oversight. When embedded proactively, these resources transform responsible AI from a compliance burden into a design advantage that improves robustness, legitimacy, and long-term trust. (NIST, 2023; European Union, 2024; Gebru et al., 2021)

## Conclusion

This analysis demonstrates that the principles of Accountability, Reliability + Safety, Privacy + Security, and Fairness that frame a portion of Responsible AI must be proactive and lifecycle-integrated, not reactive or post-hoc. Each CRISP-DM phase becomes a site for embedding these principles. Technical reliability alone is insufficient. Ethical safeguards, documentation, oversight, and regulatory alignment must accompany development from inception through deployment. We must ultimately treat Responsible AI not as a set of abstract values, but as a structured system of documented authority, risk management, fairness, and iterative monitoring across the entire model lifecycle.

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Artificial Intelligence was used for limited structural and editing support. All analysis, argumentation, and source integration were completed and verified by the authors.